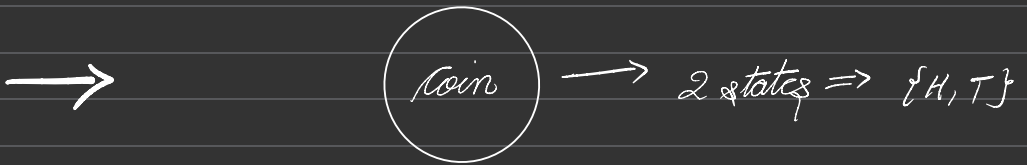


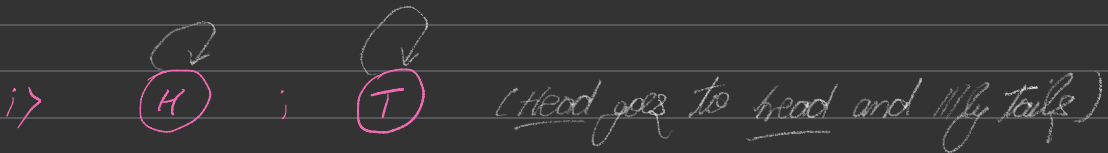
Classical Mechanics

Color-coding: Questions; Derivations; 1-line summary
Connections / Insights; General

Lecture - 1 :



Refining our own laws of motion :-

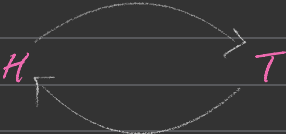


i.e. : If you start w/ H, you'll always get H and same for tails.

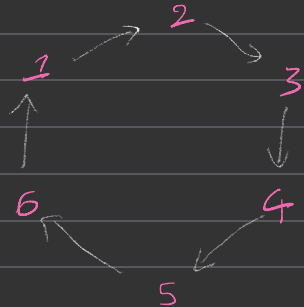
$$(H=1; T=-1)$$

Now, eqⁿ of motion for (i) :

$$\sigma(t+1) = \sigma(t) \quad (t \in \mathbb{Z})$$

ii)  $\Rightarrow \sigma(t+1) = -\sigma(t)$

$\rightarrow$ Pie $\rightarrow$ 6 states $\Rightarrow \{1, 2, 3, 4, 5, 6\}$

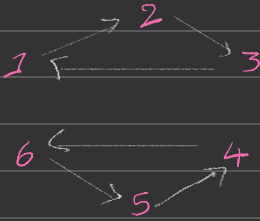
i)  $\Rightarrow \cancel{\sigma(t+1)} = 1 + \sigma(t)$

$$\Rightarrow \begin{cases} \sigma(t+1) = 1 + \sigma(t) ; \sigma(t) \leq 5 \\ \sigma(t+1) = 1 ; \sigma(t) = 6 \end{cases}$$

correct?

(2 cycles)

ii >

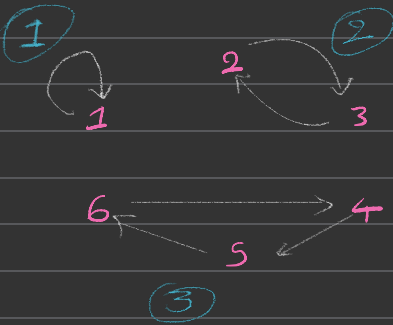


conserved quantities?

How? What does conserved q. even mean?

got it!

iii >



1, 2, 3 are conserved quantities

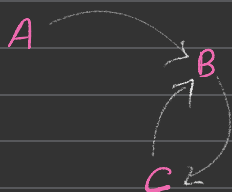
If I start w/ conserved quantity 2, I will stay on 2, even tho the state may change, 2 remains conserved.

Like 2 being mechanical energy and kinetic, potential energy being 2 states of that, interchangeable, but mech. energy remains conserved.

→ Law not allowed in classical mech.

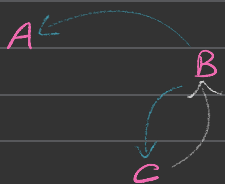
Law that you can't trace back to past.

eg :



completely deterministic law
(for future)

but if you reverse it to go back in past :



'B' becomes pt. of unpredictability
(undeterministic)

i.e. it's not reversible !!

∴ not allowed.

→ Principle of least action

(stationary)

for 1D

$$\frac{dV(x)}{dx} = 0$$



$$\frac{dV}{dx} \delta x = \delta V$$

($F_{\text{net}} = 0$)
(equil.)

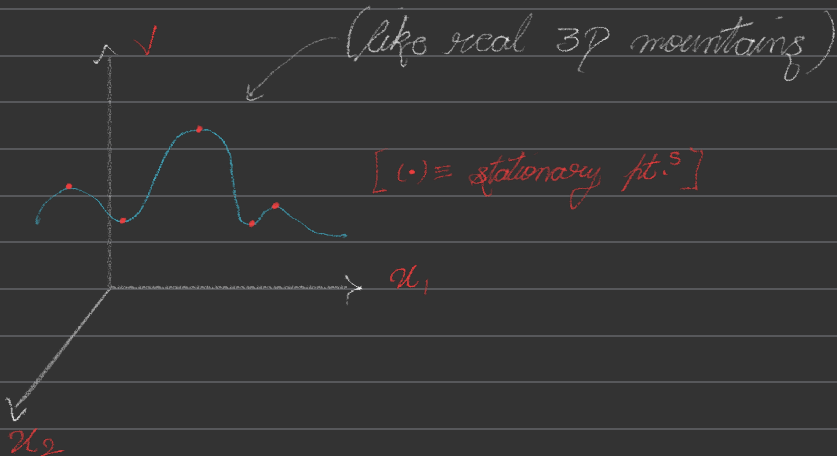
for $\geq 1D$

$$\frac{\partial V(x)}{\partial x_i} = 0$$

change in V per unit x
times small change in x

i.e. δx is the small

change in V , i.e. δV





$(\bullet) \equiv$ saddle / stationary pt.

$$\text{Here, } \delta V = \frac{\partial V}{\partial x_1} \delta x_1 + \frac{\partial V}{\partial x_2} \delta x_2$$

$\therefore$ when at equll. $\delta V = 0$